A Simplified Framework for Sequencing of Transportation Projects Considering User Costs and Benefits

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ABSTRACT

City, regional and state transportation planning authorities typically need to undertake multiple transportation improvement projects to maintain the desired level of service in a road network. Due to budgetary restrictions and to avoid inconvenience to network users, sequencing (or stage construction) of highway projects are usually considered by these decision making agencies. This paper proposes a two-step process to determine the optimal sequencing of road network capacity improvement projects. In the first step, bi-level problem determines an optimal set of capacity improvement projects for potential links in a network. Then in the second step, sequencing of projects is determined through an optimization framework with twofold prioritization criteria. The implementation feasibility of the proposed method is demonstrated through the numerical experiments using city, county and regional networks. The results of numerical experiments suggest that a proper sequencing of transportation projects can lead to significant network user travel time savings during construction.

Keywords: Sequencing of projects, Capacity improvement, Project prioritization

INTRODUCTION

State of good-repair is getting significant attention as the need to preserve the transportation infrastructure has become critical in the era of constrained budget. City, regional, and state transportation planning authorities constantly undertake efforts to maintain the transportation infrastructure in good condition so that it meets the need of travelers and achieves the desired performance measures such as congestion, safety, mobility, and air-quality among others (USDOT, 2012). To achieve such objectives transportation network performance is constantly assessed and network improvements are planned ahead in time to maintain the desired level of service (LOS). Short-term planning periods range from three to five years, whereas long term periods are of 20 to 30 years. In these planning periods various projects are selected based on performance measures target set forth by the planning authorities. Typically the project selection and its sequencing in practice is done based on engineering judgment or the needs set forth by the local citizens. Generally, planning authorities analyze one project at a time, and assess its economic benefits and justification of investment. However, in a transportation system framework, it may not be prudent to analyze one project at a time by cost-benefit analysis or by expert judgment. A more customized approach will entail a comprehensive approach in which project selection and its sequencing should be done through an optimization framework that simultaneously considers multiple competing projects in a given network.

In view of high cost of capital and growth rates of traffic, considerations of sequencing (or stage construction) of project is imperative. Stage construction is defined as a process of progressive improvements in the capacity and/or quality of a transport facility over time, beginning with a relatively low cost improvement. However, even a high cost improvement projects should get higher priority if that can bring higher benefits to network users. This entails an analytic process for the selection of the optimal sequencing and the establishment of the optimal times for implementing each project over the planning horizon. However, such procedures and analytical approaches are not often considered in practice.

Investment decisions for a set of road network improvement projects are typically evaluated using a bi-level framework: the leader-follower principle. The leader is a policy maker who tries to obtain optimal set of projects in a given budget to obtain maximum network performance measure. Network users are followers and they respond (by travelling on appropriate paths) to the strategy set by the leader. The leader-follower principle is well established in transportation research. The bi-level framework is typically used to identify the set of critical links that require the capacity expansion and the extent of their capacity expansion to achieve the desired performance measures and LOS for the network. However, this framework cannot be used directly for deciding optimal sequencing of capacity expansion of the critical links. In practice, capacity expansion projects of all critical links cannot be taken at once because of multiple reasons. First, it will necessitate substantial funding. Second, it will require the rerouting of large number of network users. Third, it may result in significant delay or increase in system travel time due to creation of multiple construction zones. This leads to an important research problem, “How to determine the optimal sequencing of the capacity expansion projects for a network, given the set of capacity expansion projects that needs to be implemented over the planning horizon”?

There are two aspects of this problem. First, the project that leads to higher system level user benefits should be implemented first to maximize the accrued benefits over time. Second, the system level cost to users during the construction period should be minimized over the planning horizon. Here user benefit is defined as the travel time savings after the completion of a capacity improvement project. The cost to user is defined as the increase in travel time due to creation of construction zone during the implementation of a project. As the user costs (as well as benefits) of the capacity expansion projects are typically interdependent, the costs (as well as benefits) of a capacity expansion project will depend on set of links selected for capacity expansion and their sequence. This research consider the interdependency in project selection but not in sequencing.

However, it is captured partially in sequencing problem as implementation of projects incurring lower costs to users before implementing the projects incurring higher costs to users can decrease the total system level user cost over the planning horizon. This is because, the capacity enhancements of links by prior implemented projects may potentially provide alternative routes to network users at later stages for whom the travel time is otherwise increased due to creation of construction zone. The benefit by cost ratio adopted in past (Vickerman, 2000) may not be a perfect metric for prioritization due to multiple reasons; first, it fails to account for the system level user benefits accrued over time, second it does not provide the flexibility to assign variable importance to the user costs and benefits. Although a rich literature on this topic exists (see table 1), past studies did not simultaneously consider both, the user costs as well as the user benefits of all suggested projects over the planning horizon for deciding the optimal sequencing of projects. This study aims to bridge this gap and proposes a mathematical programing (MP) formulation that endeavors to capture both the user benefits post construction as well as user cost during construction in deciding the optimal sequencing of projects.

The proposed framework consists of two steps. In the first step, bi-level problem determines an optimal set of capacity improvements projects for potential links in a network. Then in the second step, sequencing (both sequence and time) of projects is determined through an optimization framework with twofold prioritization criteria that tends to maximize the user benefits post construction while minimizing the user costs during construction over the planning horizon.

The contributions of this paper are multifold and are significant from both theoretical and practical perspectives. First, it proposes a MP formulation to optimally obtain sequencing of capacity expansion projects in transportation network design problems. Second, in the sequencing process user cost and benefits are accounted for both, during and after periods of construction. The proposed model provides easy way for conducting sensitivity analysis with respect to weight of user benefits over user costs on optimal scheduling. Third, the computational experiments are performed using three real world networks to demonstrate the functionality of proposed methodology and the possible sequencing of chosen construction projects. The proposed methodology is flexible to consider any number of time periods in a pre-specified planning horizon and to any temporal allocation of budget over the planning period. The proposed model is transferrable and can be implemented to any problem of staged network improvement and can be a potential tool for planners in achieving an optimal sequencing of transportation projects. The rest of the paper is organized as follows. Literature review is presented next, followed by proposed methodology to address simultaneous project selection and sequencing of transportation projects in the context of a transportation network design problem. Implementation aspects of project sequencing are discussed next. Then, in the next section real world networks of different sizes are used to showcase the applicability of the proposed two stage MP framework. Then the concluding comments and future scope of the study are presented.

LITERATURE REVIEW

This section is organized in four sub-sections. First three sub-sections briefly discuss the literature related to: (1) user costs, benefits and their importance, (2) sequencing of projects as a network design problem, and (3) optimization approaches for sequencing of transportation projects, then the summary along with the gaps in literature that acted as the motivation for this study.

User Costs, Benefits and their Importance

Researchers in the past have realized that transportation infrastructure projects are of great importance to preserve the quality of travel (Mathew and Sharma, 2009; Suwansirikul et al., 1987). In a multimodal transportation network, given a set of candidate investment projects, the process of determining which investments will be funded to meet pre-specified objectives over a planning horizon is often referred as a ‘‘capital budgeting process’’ (Martinelli, 1993). The major constraint

considered in this context is the available budget for different investment combinations such as congestion, safety, air quality, climate changes, etc. The investment decision may include project evaluation, selection, sequencing, and scheduling. For independent projects in which a project's benefit or cost is not affected by the acceptance or rejection of other projects, investment planning is relatively easy. The common project measures are ''net-present value (NPV)'', ''benefit-cost ratio'' or similar economic indicator may be used as criteria to rank the project priorities. The project sequence generated from this strategy may lead to an optimal or near optimal result. However, a more customized approach in this context will be to determine the optimal sequencing using MP. This is because project ranking using ''benefit-cost ratio'' or NPV may be not be sufficient. Consider two project one having long construction time with 10 unit user costs and benefits and second project having short construction time with 100 units cost as well benefits. In this case, neither benefit-cost ratio'' nor NPV will be effective for deciding the optimal sequencing. Thus, the project selection and scheduling problem becomes far more complex and difficult to solve. For a series of projects queued up in the investment are often implemented considering expert judgment and experience which may not lead to efficient way of reducing delay and maximizing user benefits. Construction period of each project can be mutually exclusive or inclusive. The investment-planning problem of interest is to determine which projects should be adopted and when they should be implemented so that total system level user cost could be minimized and user benefits maximized over planning period or service life under budget constraints. However, past research attempted to decide sequencing of projects to meets the trip demand (Erlenkotter, 1973).

Sequencing of Projects as a Network Design Problem

Researchers in the past has developed multiple programming techniques for selecting interdependent projects in the field of transportation as well as in the general framework of project management (Chen et al., 2015; Chow and Regan, 2011; Huang and Kuo, 2013; Li et al., 2015; Steenbrink, 1974; Wei and Schonfeld, 1993). The fundamental difference between a transportation project from others is the response of transportation network users in terms of their path choice in response to improvements. Integer programming framework of capital budgeting is the first generation of models to consider project interdependency (Johnson et al., 1985; Weingartner, 1966). In general, the existing integer programming procedures are applicable only when the number of projects is relatively small and the interdependent relations among projects are relatively simple (e.g., without higher-order interactions). Janson et al. (Janson et al., 1991) developed simplistic approaches to the selection and programming of alternative route improvements to the U.S. highway network. In their model, alternative route-improvement strategies are defined as those mutually exclusive sets of link improvements within a planning horizon. They proposed a rank-and-swap heuristic (RASH) solution procedure that accounts for the interdependent costs and benefits of route improvements. An improvement to one route could affect the benefits of improving other routes and some routes might overlap, and the authors tried to capture this phenomenon by using RASH solution procedure. The results show that RASH can generate good solutions when the highway networks are simple. However, the applicability of the RASH procedure in a complex network remains to be investigated. de Neufville, and Mori (de Neufville and Mori, 1970) proposed a dynamic programming approach to obtain optimal schedule of independent projects. A number of recent research endeavors also consider project selection in a planning period (Lo and Szeto, 2009; Mishra et al., 2016; Mishra and Welch, 2012; Sharma and Mishra, 2013; Ukkusuri and Patil, 2009). Venezia (Venezia, 1977) applied dynamic programming considering transportation project but did not consider the impact of projects on congestion or delay in the transportation network. Fernandez and Friesz (Fernandez and Friesz, 1981) analyzed the influence of demand quality interrelationship on sequencing decisions of transportation facilities.

Table 1 summarizes a list of previously published papers on multi-period transportation investment with consideration of project selection in a multi-year planning period. Although this

list may not be all inclusive, it is evident that project selection in a multi-year planning period of transportation projects is well researched in the literature, however, their applications are limited to small hypothetical test networks and feasibility of past methods in real world networks is limited. In addition, sequencing of transportation projects considering both travel time savings after the completion of project as well as delays during construction period is analyzed only by few researchers. These two gaps in literature acted as the motivation for this study and methodology proposed in this paper endeavors to bridge these gaps.

<<Table 1 Here>>

Optimization Approaches for Sequencing of Projects

Project selection and sequencing is a large combinatorial optimization problem. The solution space increases more than exponentially with the problem size (i.e., the number of projects). The existing literature on investment planning focuses on independent projects, whereas published material about interdependent projects is relatively rare. Two approaches commonly used for solving project selection and sequencing may be classified into (1) integer programming (Clark et al., 1989; Johnson et al., 1985; Weingartner, 1966) and (2) dynamic programming (Erlenkotter, 1973; Morin, 1974, 1974; Nemhauser and Ullmann, 1969; Wolsey and Nemhauser, 1999). The first inherent shortcoming of these approaches is that the interdependencies among investment projects for a series interdependent system are extremely difficult to capture in such methods. The second limitation is that, for large problems, these approaches may become inefficient or even infeasible.

Jong and Schonfeld *(17)* explored the use of genetic algorithm (GA) to optimize a project selection and scheduling problem with a 20-lock waterway network. They evaluated the project sets by adapting a simple approximation model estimated from simulation results, but with lower precision than a simulation model. Taplin and Qui (Taplin and Qui, 2001) applied GA in solving real-world problems of selecting and scheduling road construction projects. Their study coped with a series of extraneous constraints including construction sequencing requirements and financial constraints. An order-based integer vector that represents the road project sequence is used as chromosome in the proposed GA. The vector is then transformed into the corresponding construction timetable. The developed GA was applied on a road network in which a deterministic assignment model was used for evaluation of project impacts. The experimental analysis showed promising features of GAs in solving a project selection and scheduling problem that was formulated as a deterministic optimization problem. However, their study did not consider the user costs and user benefits simultaneously in the determination of optimal scheduling.

Summary of Literature Review

The literature review reveals that sequencing of transportation construction projects in network design problems is still an evolving area of research. The distinct gap in literature is that user behavior is considered only in project selection but typically not while analyzing sequencing of projects. Rather in literature many studies have used simplistic benefit cost analysis or NPV method to assess the effectiveness of transportation construction project sequencing. Only few studies considered MP framework for project sequencing but their applicability to real scale network is questionable even though some researchers proposed joint problem for project selection and sequencing. The solution approaches for project sequencing proposed in past cannot implemented for large scale networks. The problem described in this paper has four dimensions. First, the set of improvement projects needs to be determined, given the travel demand in order to achieve the optimal value of a specific objective function. Second, these improvements must consider user behavior in terms of their path choice leading to new traffic equilibrium (user equilibrium). The selected projects must be sequenced in such a way that total user delay (or similar quality measure) is minimized. Third, the selected projects need to be attributed with appropriate cost of construction and time required to complete the project construction. Fourth, selected project should be

sequenced to maximize the user benefits over the planning horizon or service life of projects while reducing user discomfort during construction phases. But past studies did not consider all of these dimensions in project sequencing and their application was limited to small test networks. In this study we present a MP framework that can be implemented for large scale network, and that considers aforesaid four dimensions to decide the optimal selection and sequencing of capacity expansion projects.

METHODOLOGY

The road transportation network improvement investment decision problem is formulated as a two-step process. *The first step* includes decision of link capacity expansions and the extent of expansions. It can be represented as a leader–follower game where the transport planner makes network planning decisions, which can influence, but cannot control the users' route choice behavior. The users make their route choice decisions in a user optimal manner. This game can be formulated as a bi-level programming model, where the upper-level problem is to determine the optimal number of lanes to be added to each link in a given set of candidate links, in order to minimize total system travel cost, subject to a given budget limit, and the lower-level problem represents a user equilibrium traffic assignment problem (UETAP) that describes users' route choice behavior. *The second step* involves the sequencing of capacity expansion projects and is the focus of this study. It includes determination of optimal sequence of capacity expansion projects in such a way that the total benefit is maximized (gained from user cost savings accrued over the planning horizon) and cost is minimized (by prioritizing the low user cost projects at earlier stages that mitigate the effect of reduced capacity of network during construction at later periods). Here it is imperative to mention two important aspects of proposed methodology. First, the interdependency of benefits of expansion projects are captured directly in selection of projects (set of links and scale for expansion) in first step only and in second step sequencing of selected projects are determined using project specific user costs (during construction) and benefits (post-construction) only. This helps to decrease the complexity of problem and facilitates applicability to real-world scale networks. Second, this study uses the planning horizon as the time line for analysis and not the service life. This is based on the assumption that all projects have nearly same service life which is much higher than planning horizon, and therefore, the net user benefits of the projects remain the same after planning horizon (as all projects are completed) till service life irrespective of their order of implementation. Notations used throughout the paper are shown below.

Network

a	Link id
A	Set of links
$\overline{A}$	Set of potential links selected for improvement, $\overline{A} \subseteq A$
i	Number of lanes to be added (0, 1, 2)
I_ϕ	Set of number of lanes to be added to link $\phi \in \overline{A}$. For example, $Ia = \{0, 1, 2\}$ means
i^ϕ	Number of lanes added in link ϕ
l^ϕ	Length of link ϕ

Flow

v_a	Flow on link a, $a \in A$

y_a^i	Binary decision variable which equals 1 if and only if *i* lanes ($i \in I_a/\{0\}$) are added to link, $a \in \overline{A}$,
t_a	Travel time on link *a*
f_r^w	Flow on path $r \in R_w$, and O-D pair $w \in W$
q_w	Given travel demand for O-D pair $w \in W$
v_a^ϕ	Flow on link *a* when capacity of link ϕ is increased after completion of construction
$\tilde{v}_a^\phi$	Flow on link *a* when capacity of link ϕ is reduced due to construction zone creation

Cost

B	Total budget for the planning horizon
b_p	Budget for time period *p*, and B $= \sum_p b_p$
μ	Construction cost of capacity expansion per unit length per lane
σ_ϕ^i	Construction cost of adding *i* lanes to link $\phi \in \overline{A}$, $i \in I_\phi$, σ_ϕ^i = 0 if *i*=0
π_p^ϕ	Cost of construction on link ϕ during period *p*
K_c^ϕ	Total cost to users during construction on link ϕ
K_b^ϕ	Total benefit to users after construction of link ϕ
t_a	Travel time on link *a*
t_a^ϕ	Travel time of link *a* after completion of construction on link ϕ
$\tilde{t}_a^\phi$	Travel time of link *a* during construction on link ϕ

Others

T	Planning horizon
p	Time period of construction (discrete time step)
θ	Construction time per unit length per lane of a link
τ_{con}^ϕ	Construction period for link ϕ
x^ϕ	Time at which the construction project of capacity expansion of link ϕ starts
τ_s^ϕ	Time interval when construction on link ϕ starts, $\tau_s^\phi = \text{ceil}(x^\phi)$
τ_e^ϕ	Time interval when construction on link ϕ ends
λ	Relative weight of benefits over cost to users

As mentioned before ***first step*** includes selection of potential links for capacity expansions and the extent of capacity expansions. The literature in this domain shows many studies for this step (Abdulaal and LeBlanc, 1979; Chootinan et al., 2005; Fisk, 1984; LeBlanc and Boyce, 1986; Wang et al., 2013; Yang and H. Bell, 1998) and this is not the main focus of this study but it is included here for completeness. The *first step* is typically formulated as bi-level problem as described next.

Step-1A: Upper Level (Decision Maker's Preference)

The objective of the decision maker is to minimize total system travel time in the network. The objective function is shown in equation (1). The objective function consists of two terms. The first term refers to the links not selected for improvement. The second term represents link with potential for improvement. The flow (v_a) is obtained from the lower level (UETAP). Equation (2) ensures that exactly one capacity level (zero, one or two lanes) is chosen for each candidate link. Constraint (3) ensures that the total construction cost does not exceed the available budget. The improvement decision variable y_a^i is binary and shown in equation (4).

$$\min Z1 = \sum_{a \in A/\overline{A}} v_a t_a(v_a) + \sum_{a \in \overline{A}} \sum_{i \in I_a} y_a^i v_a t_a^i(v_a) \tag{1}$$

s.t.

$$\sum_{i \in I_a} y_a^i = 1, \forall\, a \in \overline{A} \tag{2}$$

$$\sum_{a \in \overline{A}} \sum_{i \in I_a} \sigma_a^i y_a^i \leq B \tag{3}$$

$$y_a^i \in \{0,1\}, \qquad \forall a \in \overline{A}, i \in I_a \tag{4}$$

Step-1B: Lower Level (User Behavior)

The lower level consists of UETAP with fixed demand and is represented by equations (5) through (8) (Sheffi, 1985). Equation (5) represents the objective function of UETAP. Constraint (6) defines the demand conservation. Constraint (7) defines the relation between link flow and path flow. Constraint (8) represents the non-negativity of path flows.

$$\min\ Z2 = \sum_{a \in A/\overline{A}} \int_0^{v_a} t_a(x)dx + \sum_{a \in \overline{A}} \sum_{i \in I_a} y_a^i \int_0^{v_a} t_a^i(x)dx \tag{5}$$

s.t.

$$\sum_{r \in R_w} f_r^w = q_w, \forall\, w \in W \tag{6}$$

$$v_a = \sum_{w \in W} \sum_{r \in R_w} \gamma_{ar}^w f_r^w\ , \forall\, a \in A \tag{7}$$

$$f_r^w \geq 0, \forall\, r \in R_w, w \in W \tag{8}$$

Step-1A and Step-1B in combination determines the optimal number of lanes that needs to be added to each candidate link in a road transportation network. Wang et al. (Wang et al., 2013) provides a detailed formulation and solution approach to analyze capacity expansion problems using Step-1A and Step-1B.

Once the optimal capacity expansion projects are determined the next task is to determine the optimal sequencing of expansion projects (each project here implies capacity expansion of a link). The sequencing problem (Step 2) is the focus of this study and is presented next.

Step-2: Sequencing (Chronological staging of link capacity expansion projects)

This study considers twofold prioritization criterion to obtain optimal sequencing. The first criterion provides a higher priority to a project that leads to higher user benefits post construction. The second criterion assigns a lower priority to a project that incurs higher cost to users during construction. Based on these criteria an MP problem is formulated and is represented by Equations (9)–(13). The objective function in (9) consists of two parts: first part represents maximization of accrued user benefits over the planning horizon and second part represents impediment to projects (by delaying the start time of construction) that incur higher user costs during construction. The objective function contains a weight parameter (λ) as a tradeoff between benefits and costs. The cost and benefit terms are explained later in this section. The maximization objective function (9) can be represented by equivalent minimization objective function (9a) or (9b) utilizing the fact that MP is not affected by addition or removal of constant terms. Here onwards we will use (9b) due to its intuitiveness (evident from its force analogy presented later). Constraint (10) is to ensure that the consumed budget (sum of π_p^ϕ) in a time period p is less than the available budget (b_p). Constraint (11) ensures completion of the suggested projects in the planning horizon (T). Constraint (12) ensures that construction starts between the first and the last period during the planning horizon. Equation (13) represents non-negativity constraint.

$$\max\ Z3 = \lambda \sum_{\phi \in \overline{A}} K_b^\phi \left(T - x^\phi - \tau_{con}^\phi\right) + (1-\lambda) \sum_{\phi \in \overline{A}} K_c^\phi x^\phi \tag{9}$$

$$\text{or}, \min\ Z4 = \lambda \sum_{\phi \in \overline{A}} K_b^\phi x^\phi - (1-\lambda) \sum_{\phi \in \overline{A}} K_c^\phi x^\phi \tag{9a}$$

$$\text{or}, \min\ Z5 = \lambda \sum_{\phi \in \overline{A}} K_b^\phi x^\phi + (1-\lambda) \sum_{\phi \in \overline{A}} K_c^\phi \left(T - x^\phi - \tau_{con}^\phi\right) \tag{9b}$$

s.t.

$$\sum_{\phi} \pi_p^\phi \le b_p, \forall p \in (0, T) \tag{10}$$

$$T - x^\phi - \tau_{con}^\phi \ge 0\ , \forall \phi \in \overline{A} \tag{11}$$

$$1 \le x^\phi \le T - 1 \tag{12}$$

$$\pi_p^\phi \ge 0\ , \forall \phi \in \overline{A}, p \in (0, T) \tag{13}$$

In the proposed sequencing formulation represented by Equations (9b)–(13), the set of decision variables are represented by the vector $\{x^\phi\}, \phi \in \overline{A}$. The constant part of benefit and cost terms of *Z4* are given by equation (14), and (15) respectively. The benefit associated with the capacity expansion of link ϕ is given as the difference between the total user costs before and after the construction of link ϕ as shown in equation (14). The first term in equation (14) represents total system level user cost before increase in the capacity of link ϕ and the second term represents total system level user cost after increase in capacity of the link ϕ. Similarly, the expression of cost in

equation (15) refers to the difference between total user costs during and before construction. Note that based on these definitions, the cost and benefit associated with a project remains constant irrespective of the order of their implementation. Since both benefit and cost are in units of time there is no need for considering the discounting factor in the analysis.

$$K_b^{\phi} = \sum_{a \in A} v_a t_a - \sum_{a \in A} v_a^{\phi} t_a^{\phi} \tag{14}$$

$$K_c^{\phi} = \sum_{a \in A} \tilde{v}_a^{\phi} \tilde{t}_a^{\phi} - \sum_{a \in A} v_a t_a \tag{15}$$

The construction parameters used in sequencing of projects are given by equations (16)–(19). Equation (16) represents the construction time as the function of link length (l^{ϕ}), time required per unit length per lane (θ), and the number of lanes (i^{ϕ}) to be added. Equation (17) represents the construction cost (σ_{ϕ}^{i}) as a function of link length, cost of capacity improvement per unit length per lane (μ), and number of lane to be added for a link ϕ. Equation (18) shows the cost of construction associated with a link ϕ during a single time period *p*. The operator *ceil* in Equation (18) represents the smallest following integer. Equation (18) is based on the assumption that construction cost is equally distributed among the construction periods over the planning horizon.

$$\tau_{con}^{\phi} = \theta * l^{\phi} * i^{\phi} \tag{16}$$

$$\sigma_{\phi}^{i} = \mu * l^{\phi} * i^{\phi} \tag{17}$$

$$\pi_p^{\phi} = \sigma_{\phi}^{i} / \mathrm{ceil}(\tau_{con}^{\phi}) \tag{18}$$

Here it is important to mention that this study considers only the planning horizon (T) and not the whole service life in the proposed MP formulation. This is due to fact that after completion of construction of all projects, there would not be any variation in link travel times (assuming network remains in UE) irrespective of the order of execution of capacity expansion projects owing to the uniqueness property of link flow solution of UETAP. Therefore, variation in user cost or benefits will occur till the projects are being constructed. As per the proposed formulation construction of all projects is completed within the planning horizon T. In addition, consideration of time horizon instead of service life also simplifies the formulation. Next a force analogy for the proposed MP formulation is presented that helps in depicting the proposed MP formulation through visualization.

Analogy of Proposed Sequencing Formulation

Fig. 1(a) shows the graphical illustration of the accrual of costs and benefits as a result of scheduling of a construction project. In Fig. 1(a) the start and the end point of planning horizon is denoted as 0 and *T* (years) respectively. Each year is divided into four time periods where each quarter of the year represents one time period. Let us consider a link ϕ is suggested for improvement, and the start of the construction period τ_s^{ϕ} is $P_{1,4}$ and end of the construction period τ_e^{ϕ} is $P_{2,4}$. The benefit period under the considered planning horizon is from the end of construction period τ_e^{ϕ}(in this case $P_{2,4}$) to end of planning horizon *T*. The intent of scheduling should be to maximize the accrued benefits to users over this period. If a project leads to higher user benefits post construction then its implementation time (τ_s^{ϕ}) should be moved to earlier periods (towards left in Fig 1(a)) in the planning horizon. However, if a project incurs higher cost to user during the construction period then it needs to be moved to later periods (towards right in Fig 1(a)). Here, we can think user benefit to be a welfare force trying to move τ_s^{ϕ} towards left and user cost to be inconvenience force trying to move τ_s^{ϕ} towards right. The optimal value of τ_s^{ϕ} will depend not only on user costs and benefits

of that project but its relative user costs and benefits with respect to all others suggested link capacity improvement projects.

<<Figure 1 Here>>

Here, we present a physical analogy that facilitate to understand these fictional forces (originating due to benefits and cost) of the projects and the sequencing problem more clearly. The analogy of the sequencing problem is presented with the system of elastic strings attached to blocks of magnetic material resting on a ferromagnetic metallic surface. Fig. 1(b) shows a physical analogy of the methodology proposed for staging/sequencing of projects in this study. Let's say a block of magnetic material (analogous to a link capacity expansion project with construction cost π_p^ϕ) is placed on a ferromagnetic metallic surface and tied by two elastic strings of equal length (L_0) to the fixed ends. L_0 is the distance between the end of metallic surface to the nearest fixed end. Two forces (equal to tensions in the strings) will be exerted at the two ends of block based on the changes in the lengths of elastic strings. Suppose the desired objective is to minimize the tensions in the elastic strings tied at both ends of the block. Let the left and right side forces exerted on the block in Fig. 1(b) are analogous to welfare force (of benefit) and inconvenience force (of cost) respectively. Let x^ϕ represents the elongation in the left string, and τ_{con}^ϕ (the construction period of link) is numerically equivalent to the length of block. Let the stiffness of left and right strings are K_b^ϕ and K_c^ϕ respectively. Let's further assume that the elongations remain in elastic limit. The accrued benefit is $K_b^\phi(T - x^\phi - \tau_{con}^\phi)$, the first term of the equation (9). For the accrued benefit to be higher the block should be moved towards left which is analogous to minimizing tension in left string and is represented by first term of equation (9b). Similarly allocating negative priority to higher cost project is analogous to the pulling block to the right which implies minimizing the tension in right string and is represented by second term of equation (9b). The tension in both strings cannot be minimized simultaneously. Hence, we need to assign some importance (analogous to weight λ) to each string. Fig. 1(b) shows that depending on the stiffness of strings analogous to the benefit and cost of the of an expansion project, the optimal location of block will vary. Now let's assume that the width of block is represented by π_p^ϕ. Suppose each link expansion project is represented by one such block tied with two elastic strings aligned in parallel fashion. Fig. (1b) is demonstrated only for single block (representing one link) of the network for simplicity. Fig. (1c) conceptually shows the top view of all blocks representing all expansion projects in sequencing problem. Let the magnetic field in the cross-sectional direction of a block is proportional to it width π_p^ϕ (size in that direction) and magnetic fields of blocks adds linearly in this direction. Let the constant of this proportionality is unity. The allowable limit of magnetic field in the cross-sectional direction is violated if the total magnetic field is greater than the limiting value b_p. The question remains now to obtain the optimal values of x^ϕ for all magnetic blocks so as to minimize the sum of tensions in elastic strings such that at any cross section of metallic surface sum of the widths of the magnetic blocks is less than or equal to b_p. The optimization framework represented by Equations (9b)–(13) can be used to obtain optimal location of all blocks simultaneously which is analogous to sequencing of multiple projects in a transportation network.

Next we discuss implementation aspects of the proposed methodology.

IMPLEMENTATION ASPECTS

This section deals with two important aspects related to implementation of the proposed methodology. First, the assumption made for carrying out the numerical experiments are listed then, the implementation details of the proposed methodology are explained. This study makes following simplifying assumptions for conducting the numerical experiments. These assumptions are non-restrictive and can be relaxed as per the planning preference.

(a) The planning period is considered as 5 years for Anaheim network and 7.5 years for other two test networks.
(b) Each year is further categorized into four time periods with each quarter representing one time period.
(c) The construction cost is a function of the link length and number of lanes added.
(d) The construction cost and budget corresponds to the values in the base year
(e) The construction cost is equally spread over each period during construction.
(f) The construction periods are sequential and a project once started cannot be paused and continued till completion.
(g) No construction is allowed to start in the last period of the planning horizon.
(h) Construction period for one lane mile is considered as three time periods (nine months).
(i) The link capacity is reduced to half during the construction period.
(j) Capacity improvement per lane mile is considered as $2 million.

As the focus of this study is the sequencing problem (Step-2), here we present the implementation details of this step only. The literature provides multiple methods for solving the Step-1 and can used for identification of set of links for capacity improvement. The implementation details of the proposed methodology for the project sequencing are presented using a flow chart in Fig. 2. A capacity improvement project is defined by the link id and number of lanes that needs to be added to that link. Set of projects is identified in in Step-1 and act as the input for Step-2. As illustrated by the flow chart in Fig. 2, first the UE link flows under the base condition (do-nothing scenario) is computed. Then, the first link from the set of links suggested for capacity expansion by Step-1 is selected. Corresponding this link expansion project, flows for all links of network for two situations are computed: (i) UE link flows during construction when the capacity of the link under consideration is reduced due to creation of construction zone; (ii) UE link flows when the capacity of this link is increased after the project completion (addition of lanes). Based on these two flows, the benefit term (K_b^{ϕ}) and cost term (K_c^{ϕ}) are computed using equations (14) and (15) respectively. Following the same procedure, benefits and cost terms are computed corresponding to all suggested links (capacity expansion projects). The computation of UE flows for two conditions stated above as well as the computation of benefits and cost terms for each project are independent. This provides an opportunity of using parallel computing enabling faster implementation. Once the benefit and cost terms are determined for all suggested projects, then the minimization problem represented by equations (9b)-(13) is solved to obtain the optimal sequencing of capacity improvement projects.

<<Figure 2 Here>>

NUMERICAL RESULTS

The proposed methodology is implemented using three networks of differing sizes namely, Anaheim, Chicago Sketch and Montgomery county (Maryland) networks. Anaheim city network consists of 416 nodes, 914 links, and 1,406 O-D pairs. Chicago Sketch is the abstract of Chicago regional network of relatively larger scale and contains 933 nodes, 2,950 links and 149,769 O-D pairs. Montgomery county network consist of 1,752 nodes, 4,420 links and 50,625 O-D pairs. The total link length for Anaheim, Chicago Sketch and Montgomery networks are 466, 8,196, and 2,254 miles respectively. Details of three networks are shown in Table 2. Fig. 3 (a) through (c) shows the topographical sketch of three networks. The capacity expansion is considered for 10, 20 and 40

links (suggested by stage-1) for Anaheim, Chicago sketch, and Montgomery county networks respectively.

<<Table 2 Here>>

<<Figure 3 Here>>

The optimization problem in Step-1 has been solved using pswarm algorithm in MATLAB (Vaz and Vicente, 2007, 2009), and Step-2 (which is prime focus of this study) is solved using a generalized reduced gradient algorithm (PSP, 2013, 2011). The lower level of step-1 as well as all UE flows for step-2 have been obtained using Slope-based Path Shift-propensity Algorithm (SPSA) (Kumar and Peeta, 2014).

Anaheim Network

Table 3 presents the characteristics of the 10 links in Anaheim network suggested for capacity improvement by Step-1 and considered for sequencing problem in Step-2. The first five links in the table are suggested for two lanes addition and the bottom five links are suggested for one lane addition by Step-1. Table 3 also presents the expected construction cost, the number of construction periods needed to complete the capacity expansion, cost to users during construction and user benefits after the construction of expansion projects of these links computed using the numerical assumptions mentioned earlier in the paper.

<<Table 3 Here>>

It is evident from equation (9b) that λ is a weight factor applied to the user benefits (and $(1-\lambda)$ to user costs). Selection of a reasonable value of λ will depend on number of factors such as present state of the network, tolerance of the network users for disruptions due to creation of construction zones and objectives of the planning agency. In the absence of guiding principle for selecting the value of λ, this study performs sensitivity analysis with respect to this parameter to analyze its effect on the solution outcome.

The result of sequencing of capacity expansion projects for Anaheim network corresponding to λ value of 0.95 is shown in Fig. 4. The Fig. 4 presents the result of the project sequencing problem through sequencing of links (identified for capacity expansion) using the horizontal bars in the twenty time periods of planning horizon. The length of the bar represents the relative construction cost of the project. The optimal scheduling of the projects based on the proposed method is dictated by their relative user cost and benefit terms (and weight factor (λ) as evident from Fig. 5). In particular, it is seen that link 7, 8, and 9 are scheduled to start in the first time period. Link 1 and 10 are scheduled such that their construction is carried till the last time period. Construction on links 2 through 6 are suggested during the middle time periods (between period 4 and 15). Expansion project for links constructed in the beginning time periods (link 7, 8, and 9) result in relatively higher user benefit than the user costs (see Table 3) implying completion of expansion projects for these links leads to more accrued benefits compared to user costs. In contrast for links 1 and 10, although the user benefits post construction are high, it is more than compensated by much higher user cost during construction phase relative to other links; these facts leads to expansion projects on these two links scheduled towards the end of planning horizon. This verifies that results of numerical experiments are consistent with the objectives of the mathematical formulation in Step-2.

<<Figure 4 Here>>

Fig. 5 illustrates the results of sensitivity analysis with respect to λ. Fig. 5(a), through 5(c) shows the sequencing of capacity expansion projects for Anaheim network corresponding to λ values equal to 0.9, 0.7, and 0.5 respectively. As expected sequencing of projects with higher user cost shift towards right with the decrease in value of λ. This shift is due to fact that with decreasing value of λ the proposed method provides more weight to user cost than benefit. Fig. 5(a) shows that link 7 and 8 are shifted to later periods for construction. But link 9 is still suggested for improvement in the beginning of the first time period. One noticeable consistency between Fig. 4 and Fig. 5(a) is that the links 1, and 10 have the same starting period of construction and their constructions are scheduled to finish in the last time period. Similar results are shown for these two links in Fig. 5(b), and (c). In addition, another noticeable result that deserves attention is that the construction on link 2 shift to right finishing in last period for all cases analyzed except when λ equals to 0.95. Table 2 shows that link 2 ranks the second, both in terms of user costs and benefits. Hence, it's schedule is very sensitive to the choice of λ. The purpose of demonstration of Fig. 5 (a) through (c) is to depict how the required allocation of budget and sequencing of construction on the selected links is affected by the relative weight of user benefits over the user cost in a framework where benefit received is maximized and cost to the users is minimized simultaneously.

<<Figure 5 Here>>

Fig. 6 shows the objective function (weighted sum of user benefits lost and cost to users) represented by Equation (9b) for four λ scenarios. As λ decreases the objective function increases. The results of numerical experiment depicted by Fig. 6 seem reasonable. This is because the cost term (K_c) is weighted higher than the benefit term (K_b) with the decrease in the value of λ. The insight from this variation is an additional layer of information for the planning agencies as there will be multiple project sequencing options from different weights of λ. Variation in λ reflects the decision maker's weights to benefits and costs. As the future remains uncertain in terms of budget and other policy issues, an analysis that provides a set of options is useful in analyzing sequencing of transportation projects. In addition, the results presented in Fig. 6 also suggest that if sequencing of transportation projects are considered (using adequate λ) in the network design problem significant user cost savings can be obtained.

<<Figure 6 Here>>

Chicago and Montgomery County Network

The scale of network can have impact on the sequencing pattern and can involve more complex interactions between benefits and costs among suggested projects. This study considers Chicago sketch network as the second case study for demonstrating the proposed methodology on a regional network and with larger number of capacity expansion projects. Step-1 of the algorithm resulted in 20 projects as shown in Table 4. Link 1 and 4 warrants for 20 time periods of construction. To obtain a realistic results 30 planning periods are chosen for analysis (equivalent to 7.5 years of planning horizon). The budget of $150 million is considered for Chicago network over the planning horizon. A budget of $5 million for each time period is considered (with equal spread over the planning horizon) in this case.

<<Table 4 Here>>

Sequencing of construction projects over 30 planning periods for Chicago Sketch network (using λ equal to 0.95) is shown in Fig. 7. Sensitivity analysis with respect to λ is not shown for Chicago Sketch network as similar insights were obtained by varying λ as were obtained for Anaheim network. Fig. 7 shows that high benefit links are suggested for early construction and vice

versa for low benefit links. An important insight from the numerical results presented in Fig. 7 is that projects requiring lesser time periods are shifted to left and are scheduled to start early than project requiring higher number of time periods for construction.

<<Figure 7 Here>>

The numerical experiment for sequencing of projects for Montgomery County network is performed to test the proposed methodology and draw insights on a county level network. A list of 40 projects was selected from step-1, and sequencing methodology proposed in study is applied to obtain the optimal sequencing. Input data and link characteristics of these 40 projects are not presented in the paper to preserve space. The result of numerical experiments for Montgomery county for λ equal to 0.95 is shown in Fig. 8. There are three important insights from the results presented in Fig. 8. First, the length of the construction for the project varies by large amount in this case. Second, most of the projects finishes by the mid of the planning horizon. Third, many short length projects requiring even a single time period for construction are scheduled for later periods. This is due to the fact that creation of construction zone on them even for short duration brings relatively higher costs for users compared to accrued benefits post construction.

<<Figure 8 Here>>

The average computational time per iteration of Step-1 and Step-2 is 15 minute and five minutes respectively for Chicago sketch network. The computational time for Anaheim network was found to be about one tenth of Chicago sketch network and about twice for Montgomery county network. The computation was performed using a Windows 7 64-bit operating system, with i7 Quad Core Processor and 32 GB RAM. This vindicates the computational feasibility of proposed methodology for real scale networks.

The proposed research on sequencing of transportation projects has number of practical considerations. First, the simple yet comprehensive framework allows transportation planners and program managers in scheduling the projects for construction to obtain network level savings in terms of user costs and benefits within their budget constraints. Second, the computational resources needed for using the proposed models are moderate, hence the approach allows seamless integration of sequencing of projects in transportation planning process. Third, the proposed framework incorporates increased user costs due to construction into the project sequencing problem which is largely ignored. Validation of sequencing results is challenging, however, the numerical results for various real world networks used in this study suggest that any other sequence of projects than obtained by proposed framework leads to inferior total system travel time, demonstrating that the reported results are optimal.

CONCLUSIONS AND DISCUSSIONS

This paper demonstrates sequencing of transportation projects by considering user behavior in transportation network design problems. The proposed methodology is based on a two-step process, where first step is a bi-level problem that suggests the capacity improvements for the subset of links in a network (not the focus of this study), and in the second step sequencing of projects is determined. The primal focus of this study was to develop a framework for deciding the optimal sequencing of projects that captures the relative important of projects based on twofold prioritization criteria while assigning relative weight to them: maximize the accrued benefits to users while deprioritizing the projects that entail high user cost during construction. A MP formulation is developed based on this double prioritization criteria. The implementation of proposed methodology is demonstrated using three real size test networks, namely the Anaheim, Chicago sketch and Montgomery county networks. The sequencing model provides the flexibility

to decision maker in using weights to the benefit and cost components to tackle budget and other uncertain policy measures. Multiple scenarios by varying weights are analyzed for Anaheim network. The results indicates that if the weight for benefit is high, the projects with relatively higher benefits is scheduled to start in earlier time periods even though it incurs higher cost to users during construction phase. Further, numerical results also suggest that λ plays a critical role in sequencing of projects as well as system level user cost savings.

Contribution of this study is multi-fold and is significant from both theory and practice. First, the study proposes an optimization framework for sequencing of construction projects that evaluates the relative importance of all projects simultaneously. Second, in the sequencing process user cost is accounted for both, during (user costs) and after (user benefits) periods of construction. Third, the proposed methodology is demonstrated using three real world networks of different scale: city network, regional network and county network. Fourth, the methodology is flexible to consider any number of times periods in a pre-specified planning horizon and budget distribution among these time periods. The proposed model is implemented to network of any scale and can be a potential tool for planners while considering sequencing of transportation projects. The proposed methodology did not consider interdependencies of projects in term of equilibrium flows that can increase the problem complexity as user equilibrium needs to be considered in every combination of project sequencing. A useful future direction to this study will be to incorporate these interdependencies in the sequencing problem. The study considers only user costs and benefits in the sequencing problem, while in reality planning agencies work with multiple performance measures. Nonetheless, the paper proposes a robust methodology for sequencing of projects by factoring both user costs and benefits in a multiyear planning horizon with multiple time period framework. Although, the proposed methodology is applied in network design problem context, it can be useful for other research areas involving sequencing problems under multiple prioritization criteria.

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Table 1 Literature review of multi-period investment decision making using network design problems.

Author	Multiple Periods	Project Staging	Objective	Application-Hypothetical	Application-Real World
(Lo and Szeto, 2009)	✓	✓	Max. Consumer Surplus	✓	×
(Ukkusuri and Patil, 2009)	✓	×	Max. Consumer Surplus	✓	×
(Szeto and Lo, 2005)	✓	✓	Max. Consumer Surplus	✓	×
(Chen and Subprasom, 2007)	✓	×	Multiobjective (Profit, Consumer Surplus, and Equity)	×	✓
(Chow and Regan, 2011)	✓	✓	Max. Net Present Value	×	✓
(Pan and Xu, 2013)	✓	✓	Max. Consumer Surplus	✓	×
(SUN et al., 2011)	✓	✓	Max. Consumer Surplus	✓	×
(Patil and Ukkusuri, 2011)	✓	×	Max. Consumer Surplus	✓	×
(Ye and Ukkusuri, 2015)	✓	✓	Max. recovery performance	✓	×
(Chow et al., 2011)	✓	×	Max. Net Present Value	×	✓
(Albareda-Sambola et al., 2012)	✓	×	Min. route cost	✓	×
(Luathep et al., 2011)	×	×	Min. total system travel time	✓	×
(Szimba and Rothengatter, 2012)	✓	×	Max. utility cost	✓	×
(Jiang and Szeto, 2014)	✓	✓	Max. Consumer Surplus	✓	×
(Sun and Turnquist, 2007)	×	Yes	Max. Expected Systems Capacity		
(Hsieh and Liu, 2004)	✓	Yes	Max. investment utility	✓	×

Table 2 Networks used for demonstrating the optimal sequencing of projects.

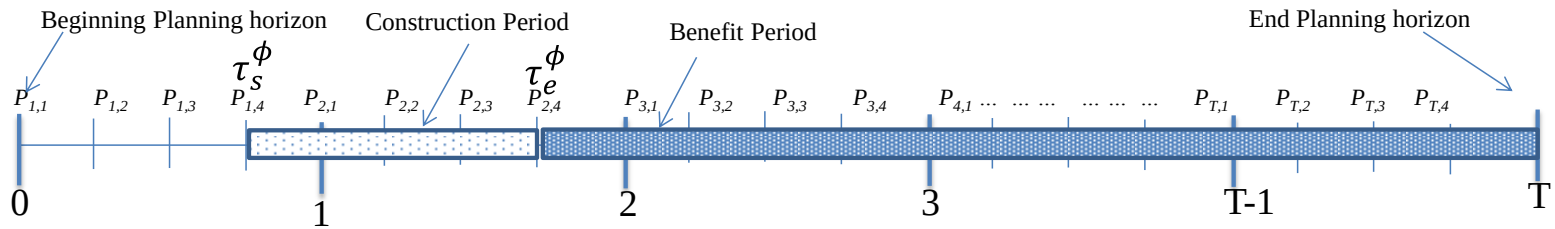
Network	Nodes	Links	Zones	O-D pairs with non-zero demand
Anaheim	416	914	38	1,416
Chicago Sketch	933	2,950	387	93,135
Montgomery	1,752	4,420	225	50,625

Table 3 Characteristics of links considered for sequencing in Anaheim network

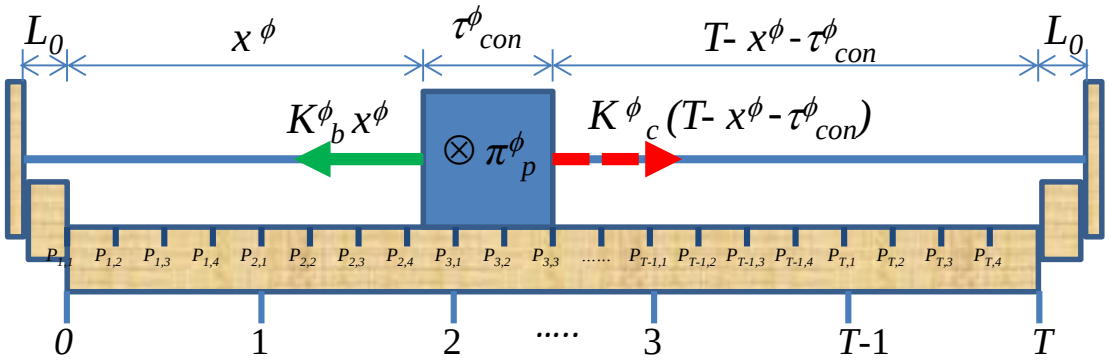
Link Number	From Node	To Node	Length (miles)	Lanes added	Expected Cost ($)	Expected Construction Periods	Cost to users during construction (K_c)	Benefit to users (K_b)
1	63	62	1.000	2	4.0	6.0	2,550,620	15,047
2	87	86	1.364	2	5.5	9.0	692,073	2,722
3	143	142	1.364	2	5.5	9.0	100,100	2,031
4	144	143	1.364	2	5.5	9.0	39,764	114
5	145	144	1.364	2	5.5	9.0	130,226	2,120
6	194	193	1.364	1	2.7	5.0	27,987	377
7	195	194	1.364	1	2.7	5.0	35,904	1,322
8	204	203	1.364	1	2.7	5.0	25,497	962
9	233	232	1.705	1	3.4	6.0	30,597	962
10	235	234	1.705	1	3.4	6.0	250,645	1,282

Table 4 Characteristics of links Considered for sequencing in Chicago sketch network

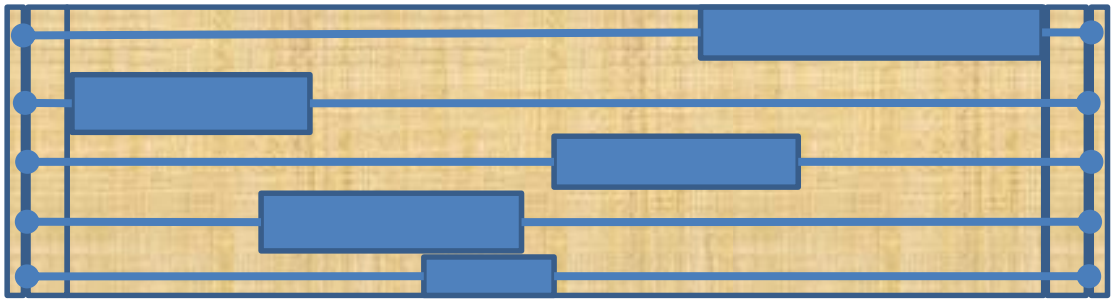
Link Number	From Node	To Node	Length (miles)	Number of lanes added	Expected Cost ($)	Expected Construction Periods
1	403	404	3.26	2	13.04	20
2	436	496	1.02	2	4.08	7
3	477	478	2.53	2	10.12	16
4	478	479	3.24	2	12.96	20
5	490	491	2.61	2	10.44	16
6	491	490	2.61	2	10.44	16
7	491	492	2.84	2	11.36	18
8	492	491	2.84	2	11.36	18
9	493	494	2.24	2	8.96	14
10	493	497	1.94	2	7.76	12
11	493	564	1.37	1	2.74	5
12	494	493	2.24	1	4.48	7
13	494	495	2.81	1	5.62	9
14	495	494	2.81	1	5.62	9
15	495	496	2.56	1	5.12	8
16	496	436	1.02	1	2.04	4
17	496	495	2.56	1	5.12	8
18	564	493	1.37	1	2.74	5
19	565	564	2.16	1	4.32	7
20	568	565	2.08	1	4.16	7



(a) Temporal dimension of sequencing problem shown for a single link



(b) Physical analogy of sequencing problem showing elevation view of a single block (single link)



(c) Physical analogy of sequencing problem showing top view of all blocks (all potential links)

Fig. 1 Force diagram analogy of sequencing problem of construction projects.

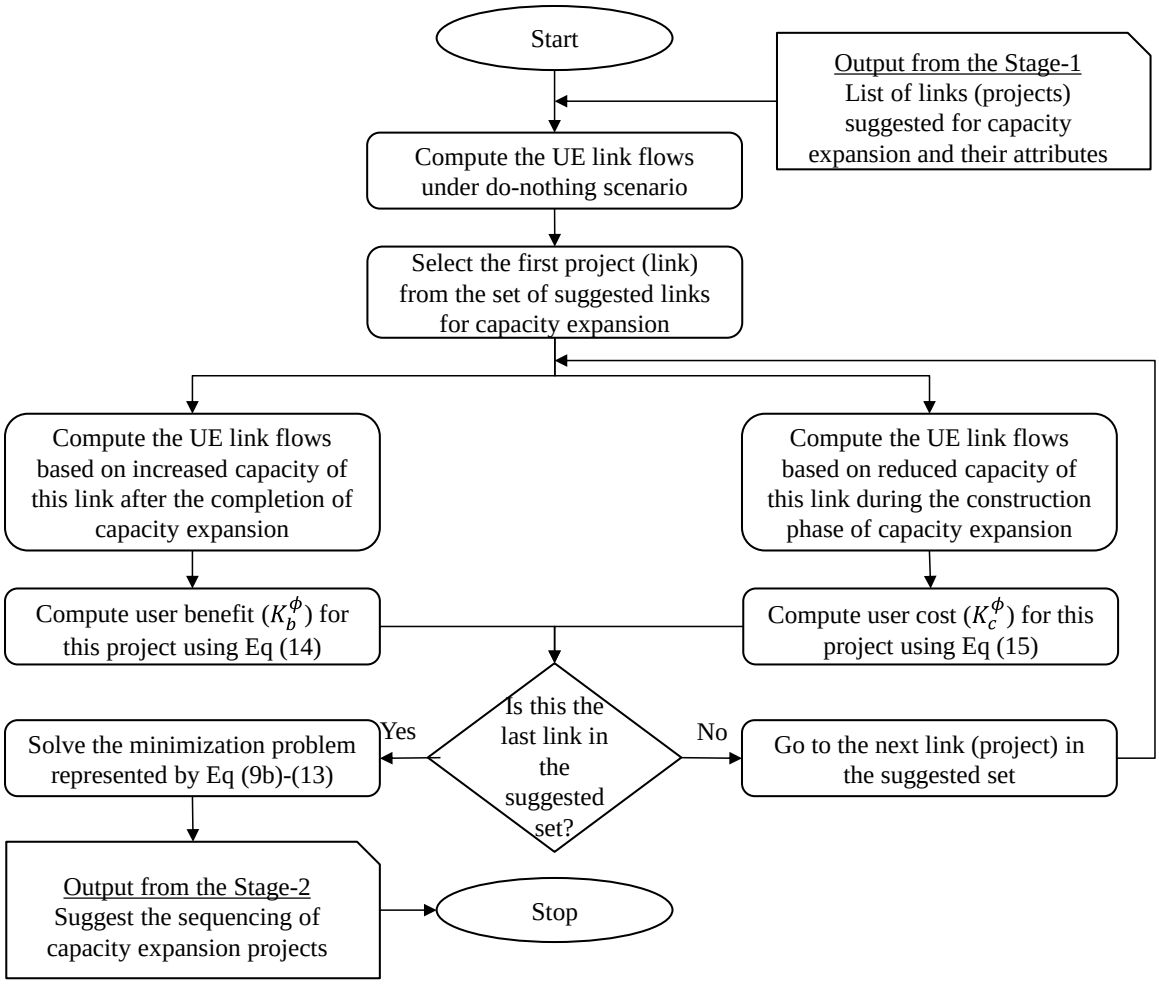
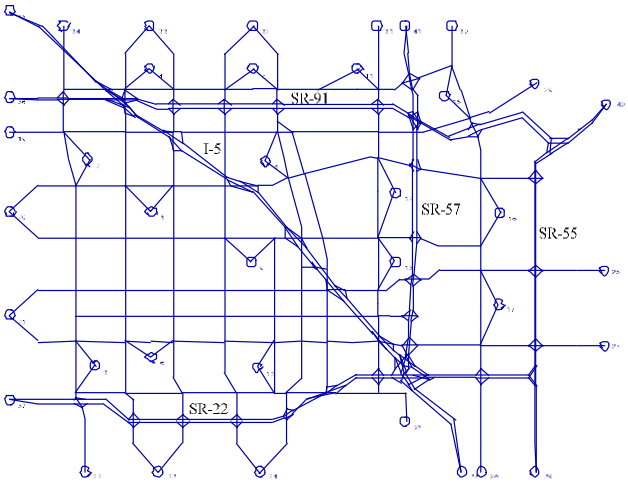
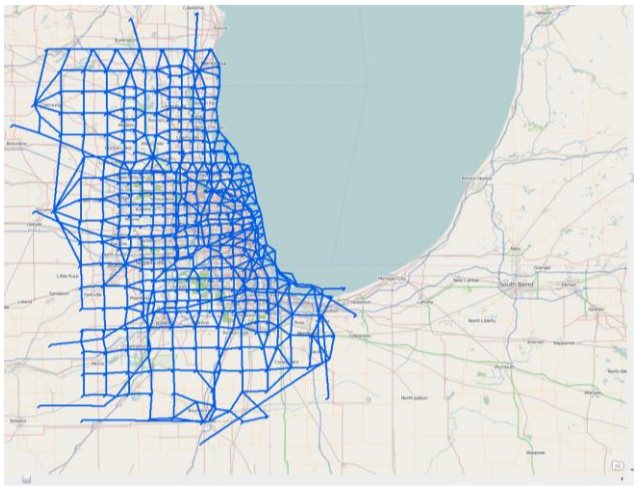


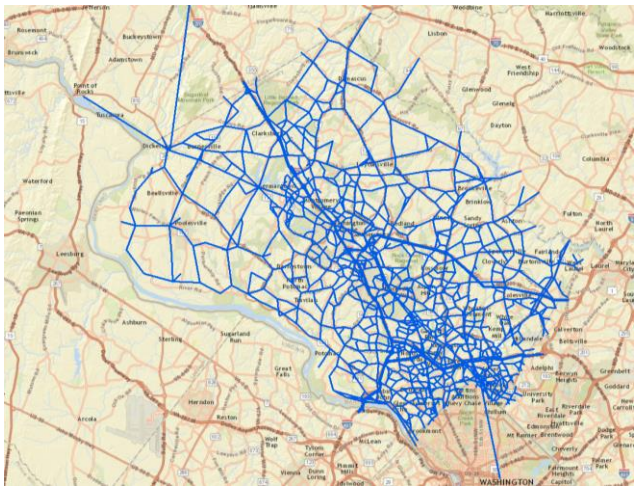
Fig. 2 Implementation details of project sequencing problem.



(a) Anaheim Network



(b) Chicago Sketch Network



(c) Montgomery County Network

Fig. 3 Topographical sketch of the analyzed transportation networks.

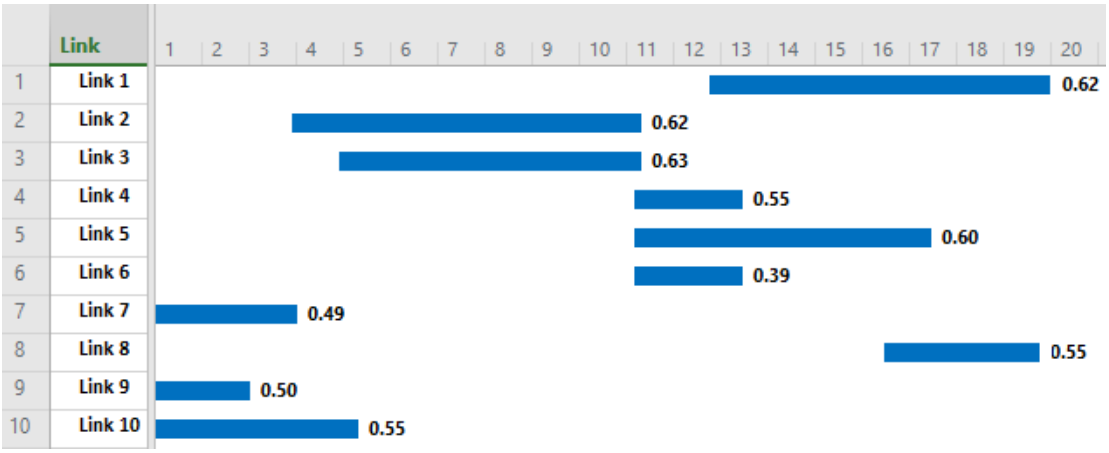
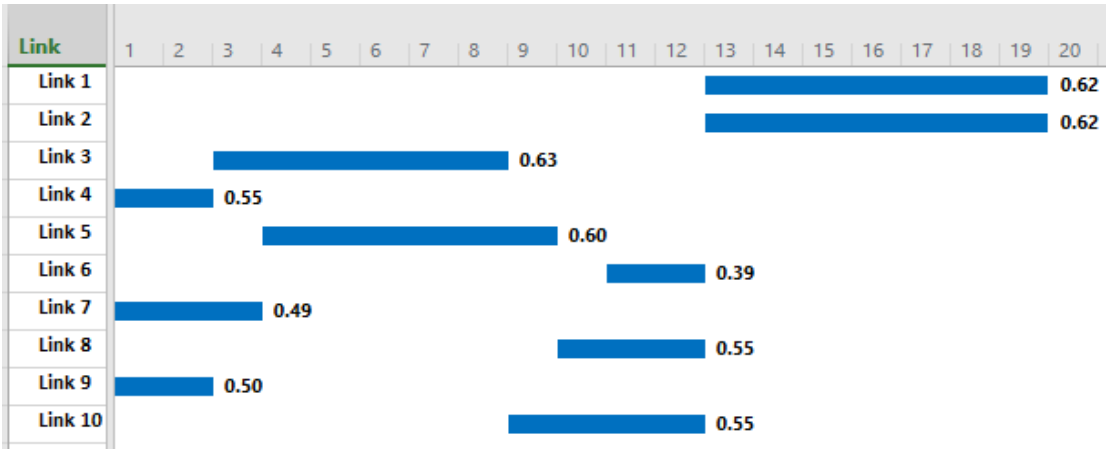
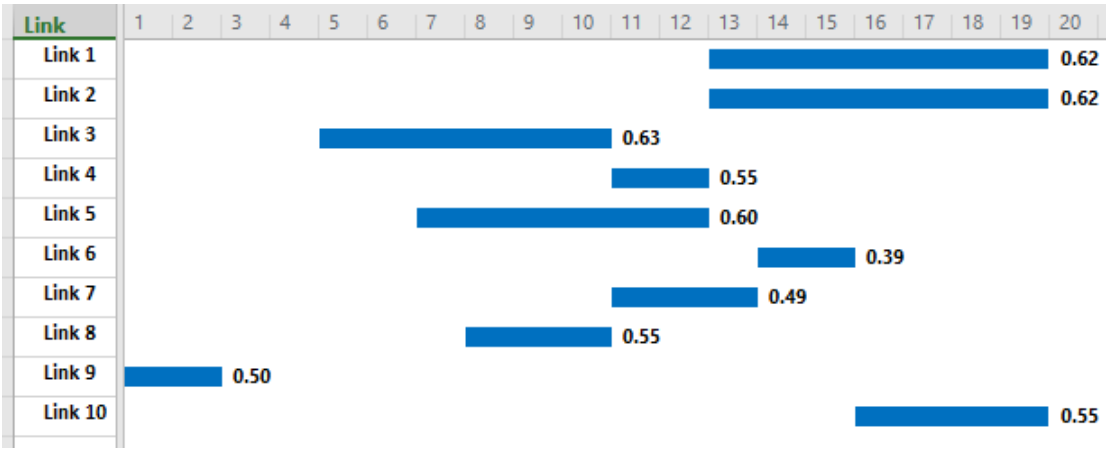


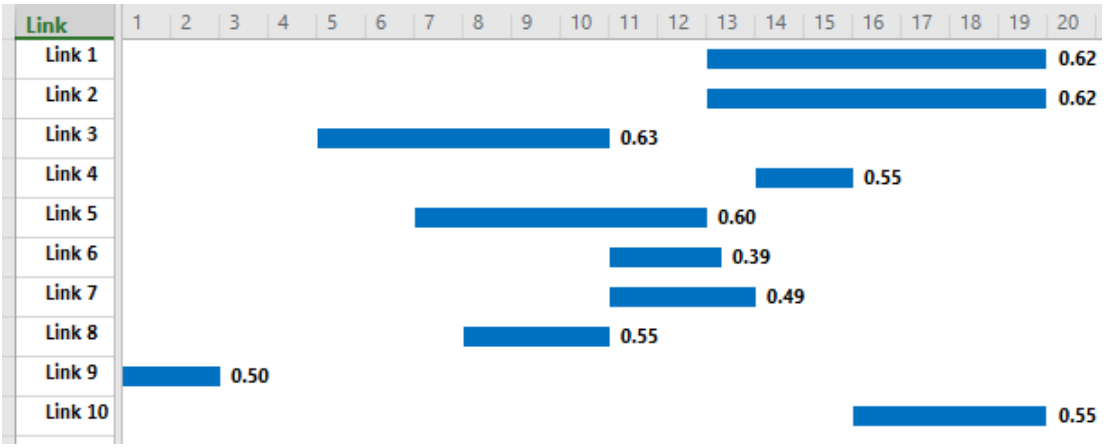
Fig. 4. Sequencing of projects for Anaheim network (λ= 0.95).



5(a) Sequencing of projects for Anaheim network (λ = 0.90)



5(b) Sequencing of projects for Anaheim network (λ = 0.70)



5(c) Sequencing of projects for Anaheim network (λ = 0.50)

Fig. 5 Sequencing of projects for Anaheim network for different values of λ.

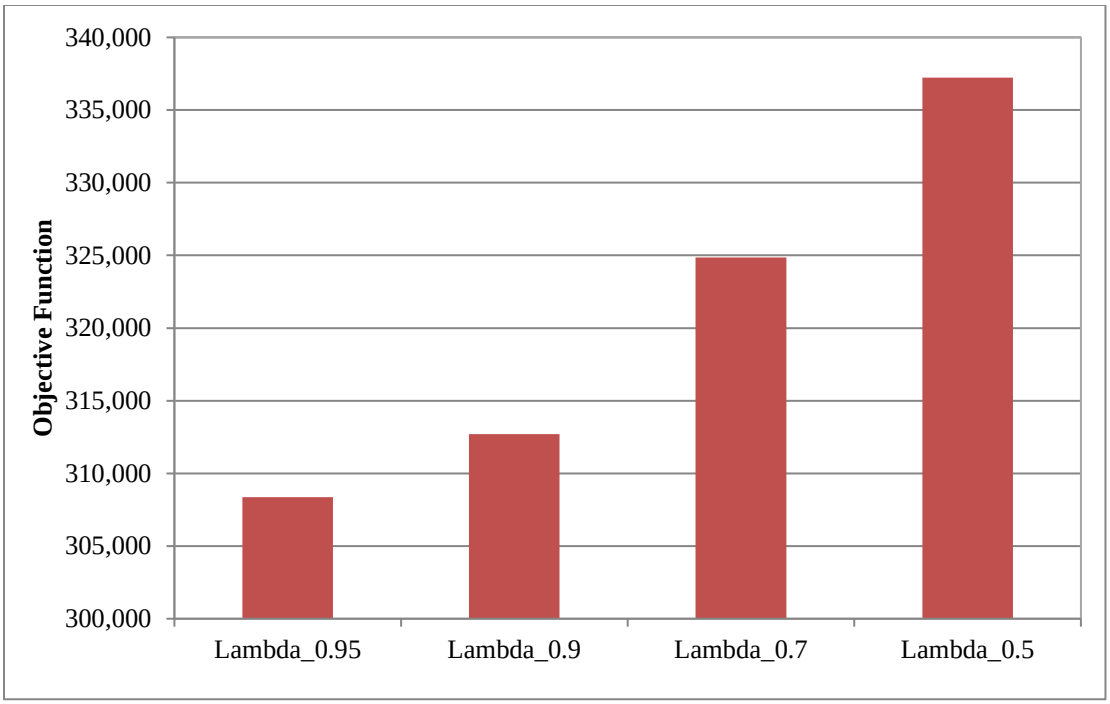


Fig. 6 Objective function values at optimality at various λ (Anaheim network).

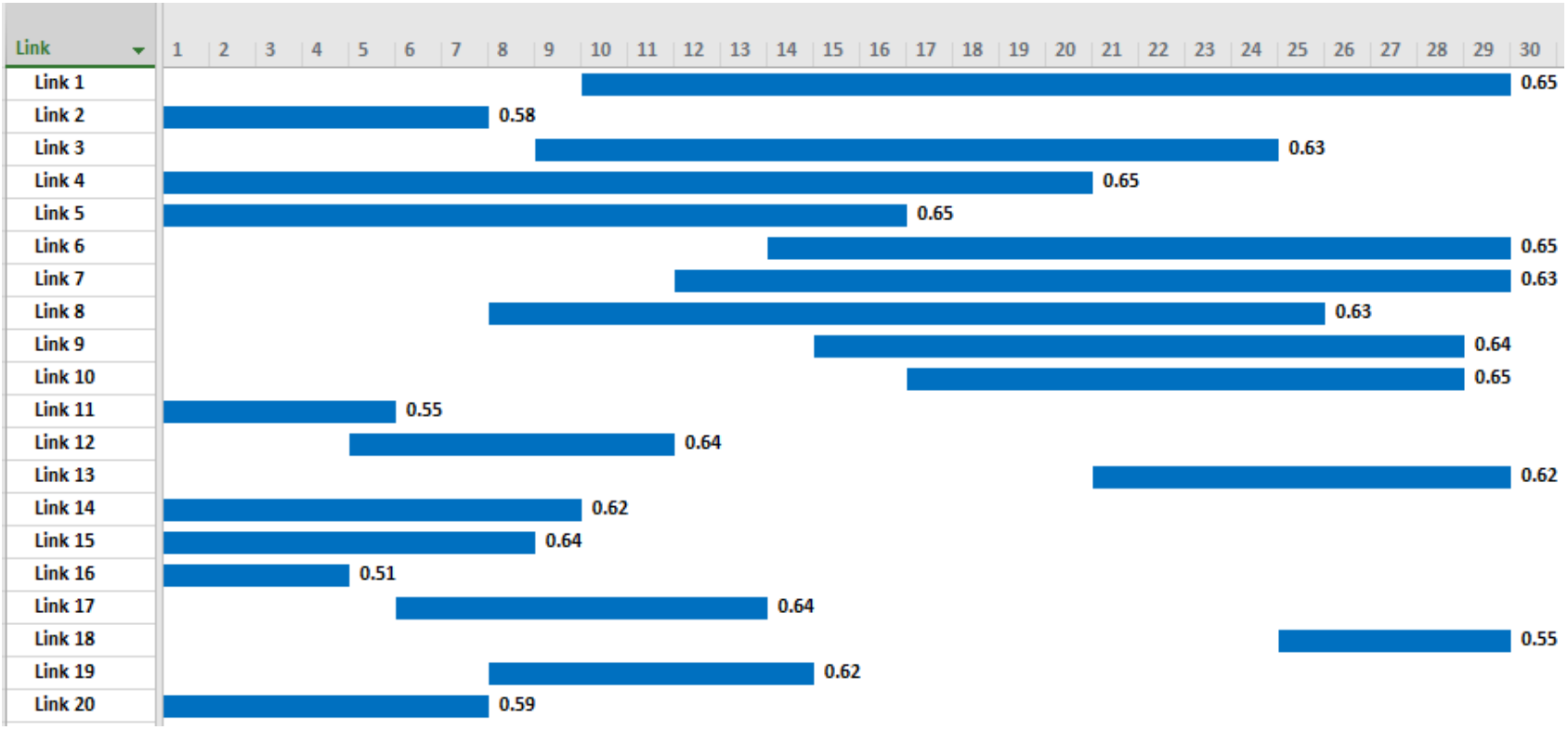


Fig. 7 Sequencing of capacity expansion projects for Chicago sketch network.

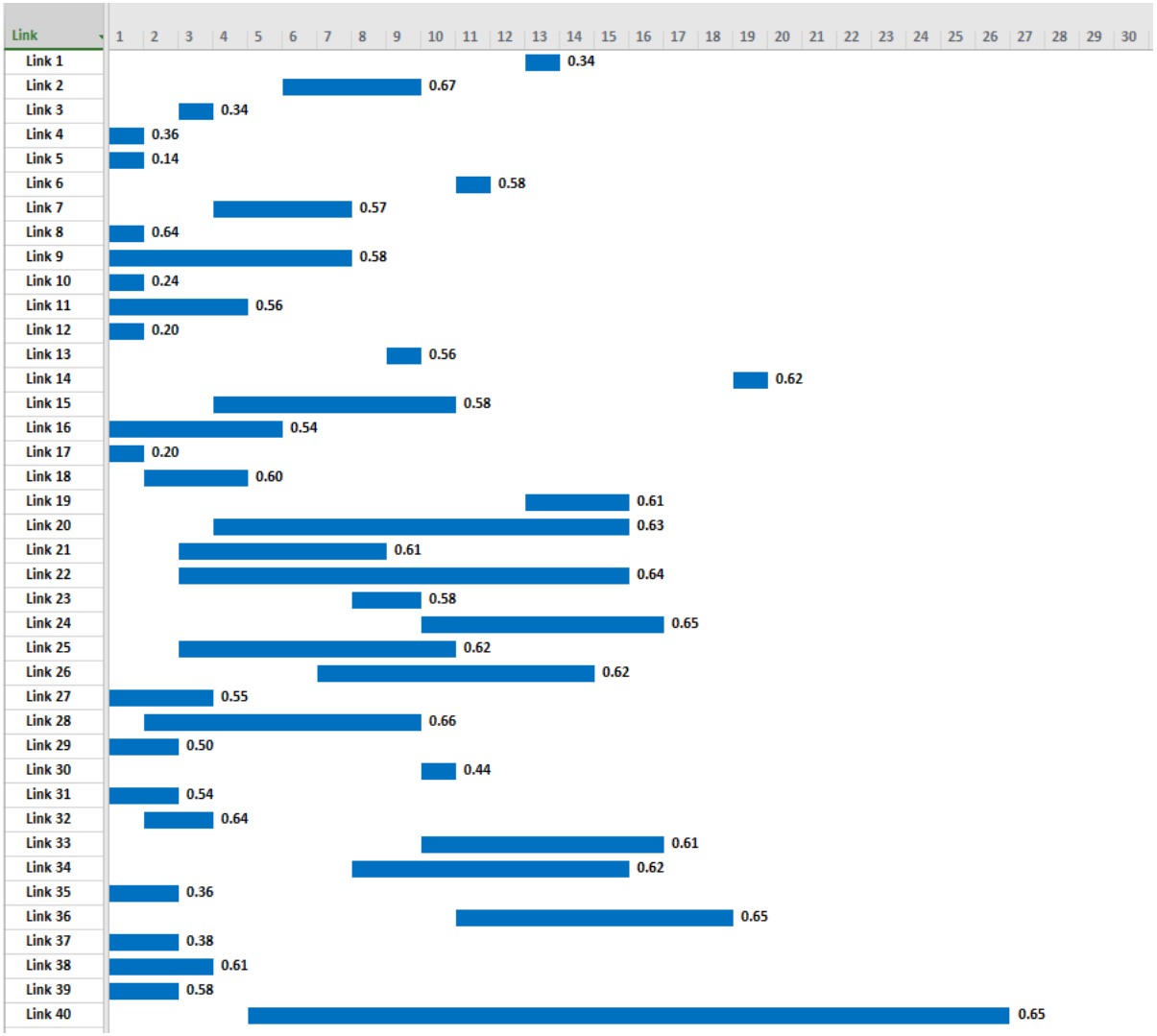


Fig. 8 Sequencing of capacity expansion projects for Montgomery county network.